Michael C.

LinkedIn: linkedin.com/in/michael-java-backend

Java Backend Engineer · Payments & Wallets · Distributed Systems

Open to remote roles

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Summary

Hands-on Java backend engineer with 13 years of experience, including Alibaba Group and DiDi, followed by wallet and payment systems at Huobi and dtcpay.

For the past 4 years I've built backend modules for wallets and payments: a non-custodial multi-chain wallet at Huobi and a consumer e-wallet at dtcpay. My focus is transaction integrity, reliability, and recovery from failures.

I take a module from design to production on my own and work well in async, remote teams across time zones.

How I Work

Written-first: requirements and decisions live in docs; async by default, meetings short and purposeful.

Pragmatic: judged by working results and data; focused on solving problems.

Professional process: code review, clear standards, clear ownership.

Evidence-led: use measurements, production observations, and technical trade-offs to resolve disagreements.

Transparent collaboration: document decisions, share context, and raise issues directly and respectfully.

Skills

Java / Spring Boot	thread pools, JVM OOM troubleshooting, transaction pitfalls
MySQL	index design, cursor pagination, conditional updates + row locks
Redis	caching, distributed locks, Bloom filters
Kafka	reliable delivery, idempotent consumers
Payments	idempotency, state machines, channel integration, reconciliation
Blockchain	wallet creation, transaction broadcast, block scanning, reorg handling

Experience

dtcpay Java Backend Engineer

Apr 2024 - Present

Singapore-based digital payments company | Remote | Website: dtcpay.com

MAS-licensed payment institution (fiat + stablecoin). Developed the core flow of the consumer e-wallet app: top-up → withdrawal → daily reconciliation; every balance change is booked by the wallet service.

Concurrency safety (no overdraft): a single conditional UPDATE checks and deducts the balance under a row lock, so concurrent debits queue up and never go negative.

Idempotent APIs: one-time Redis tokens block duplicate submits; a unique index on the business order ID is the final guard.

Lost-callback recovery: verify signature and amount on channel callbacks; scheduled jobs query the channel to recover missed notifications.

Idempotent payouts: freeze funds before payout; on channel timeout only query, never resend; deduct on success, unfreeze on failure.

Eventual consistency: debit and pending notification are written in one DB transaction, then delivered via Kafka with retries (local message table).

Huobi Senior Java Backend Engineer – Blockchain Wallet

May 2022 - Apr 2024

Global crypto brand - Huobi Wallet team | Remote | Website: www.htx.com

Huobi Wallet: a non-custodial multi-chain wallet with ~12M users (private keys stay on the user's phone). Developed the core flow: wallet creation → on-chain transfer → deposit detection → deposit notification.

Chain reorgs: detect broken parentHash links, roll back and rescan; credit deposits only after enough confirmations.

OOM fix: the block scanner crashed while catching up; root-caused an unbounded queue holding blocks in memory and switched to a bounded queue with back-pressure.

Reliable delivery + idempotent consumption with Kafka: persist notifications before sending, retry failures, and deduplicate repeated deposit events.

Address filtering with a Bloom filter: discard non-user addresses in each block before hitting the database.

Scanner HA with a Redis distributed lock: one active scanner, automatic failover, resume from the saved cursor.

DiDi Senior Java Backend Engineer — Oct 2018 - May 2022

Major global ride-hailing platform - China's Uber | Hangzhou, China | Website: www.didiglobal.com

Ride-hailing platform. Tech lead of the company-wide search service.

Built the search service from 0 to 1 on Elasticsearch.

Built SQL lineage parsing that automatically extracts upstream/downstream table dependencies, exposed as a query service.

Provided unified query APIs consumed by systems across the company.

Alibaba Group Java Backend Engineer – Distributed Job Scheduling — Oct 2016 - Oct 2018

Global e-commerce and cloud technology group | Hangzhou, China | Website: www.alibabagroup.com

Alibaba Group. Core developer of the scheduling system behind Alibaba Cloud DataWorks: master election, job sharding and timed triggers.

Master HA: nodes compete for a database row lock (SELECT ... FOR UPDATE); when the master dies the lock is released and another node takes over automatically.

Job sharding: machines map to a fixed set of logical nodes, so scaling in/out only changes the mapping (similar to consistent hashing).

Quartz-based scheduling of 1M+ hourly/daily/weekly/monthly jobs; average job development time dropped from 36h to 17.45h after launch.

Ourpalm Java Backend Engineer — Mar 2013 - Oct 2016

Publicly listed mobile game developer and publisher | Beijing, China | Website: www.ourpalm.com

Mobile game company listed on the Shenzhen ChiNext board. Built the shared server side used by multiple games: log collection SDK and operations admin.

Designed and built a server-side game log collection SDK adopted by multiple mobile games.

Developed and maintained the Java operations admin system and its scheduled jobs.

Personal Projects (AI Agent)

Homepage Personal homepage built and launched solo with AI agents (Next.js + Vercel) · michael.opc-x.org

Quarry Multi-channel data collection: scrapes job postings from BOSS, RemoteOK, V2EX, Telegram, X and more into one feed, with automatic tagging and scoring · quarry.opc-x.org

Education

Hawaii Pacific University M.S. Artificial Intelligence · GPA 3.88 / 4.0 · With Distinction — Jan 2025 - Jul 2026

University of Electronic Science and Technology of China B.Eng. Computer Science and Technology — Sep 2008 - Jul 2012